

AI UNLEASHES CHAOS

Lorenz attractor -- circuit by Paul Horowitz

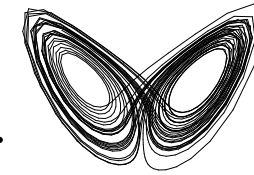
PCB by Jason Gallicchio and Claude 2026-09-15 rev A

$$\begin{aligned}dx/dt &= s(y - x) \\ dy/dt &= r x - y - x z \\ dz/dt &= x y - b z\end{aligned}$$

$$s = 10 \quad r = 28 \quad b = 8/3$$

Every term weighs 1 MEG / R: R1 =
R2 = 100k is s, R3 = 35.7k is r,
R7 = 374k is b. The MPY634s form
A*B/10 and get their minus signs
from swapped inputs. 0.1 V per unit,
time scale tau = 1 MEG x G.

x and z on a scope in X-Y draw this



this project:
github.com/gallicchio



Paul's original:
seti.harvard.edu

Sheet:
File: lorenz.kicad_pcb

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.6

Rev:

Id: 1/1